WEEK 7: Student Performance - Data Cleaning & Transformation

-Priyadharshini

E25DA033

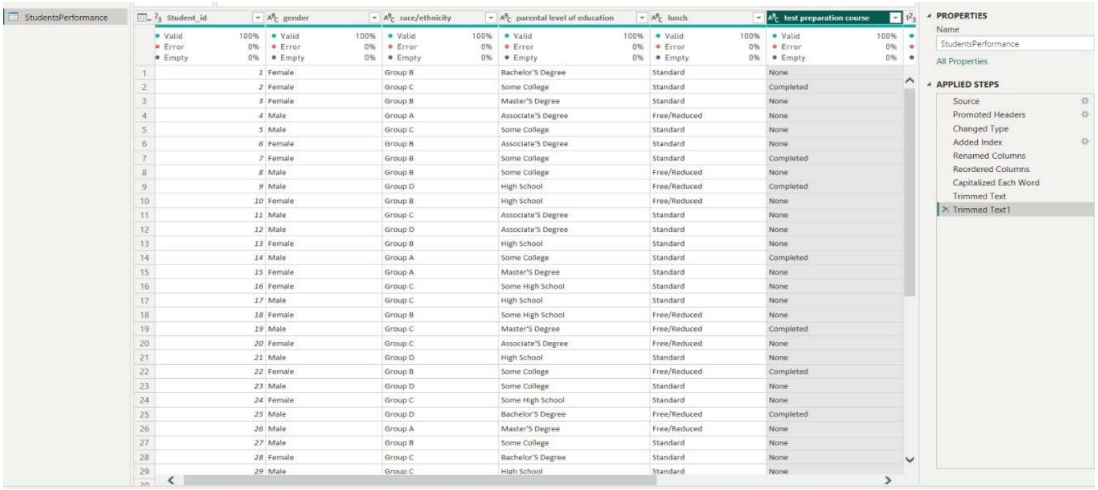
1. Executive Summary

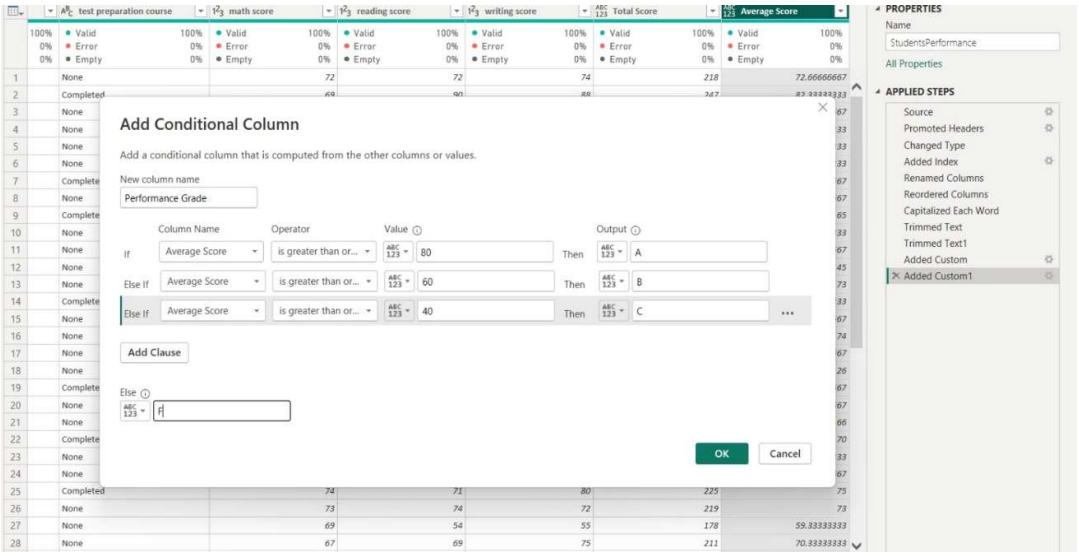
This report presents the data cleaning, transformation, and restructuring procedures performed on the Student Performance dataset using Power Query in Power BI. The main objective was to establish an analysis-ready dataset by standardizing categorical text, engineering custom performance indicators, and pivoting scores for dynamic reporting.

2. Data Cleaning & Transformation Steps

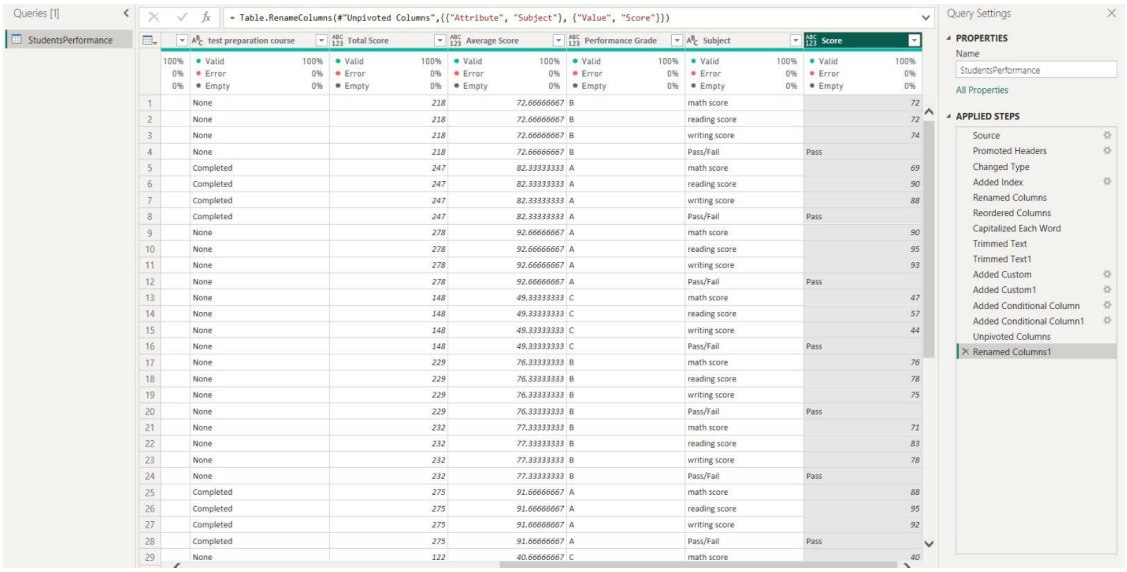
The following procedures were executed in Power Query Editor:

- **Import & Verification:** The dataset (StudentsPerformance.csv) was imported, verified for structural completeness, and confirmed to have zero null values.
- **Text Standardization:** Applied Trim and Capitalize Each Word on all categorical text fields (Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course) to fix formatting inconsistencies.
- **Primary Key Creation**: Generated an index column starting from 1, renamed it to Student_Id, and moved it to the first column position.





- **Feature Engineering:**
  - **Total Score:** Calculated using [math score] + [reading score] + [writing score].
  - **Average Score:** Computed via [Total Score] / 3.
  - **Performance Grade**: Configured using conditional column rules (\ge 80 \rightarrow \mathbf{A}, \ge 60 \rightarrow \mathbf{B}, \ge 40 \rightarrow \mathbf{C}, Else \rightarrow \mathbf{F}).
  - **Pass/Fail Classification:** Derived using conditional rules where Average Score \ge 40 \rightarrow \mathbf{Pass}, otherwise \mathbf{Fail}.
  - **Data Reshaping:** Selected the three individual score columns (math score, reading score, writing score) and performed Unpivot Columns, converting them into structured Subject and Score fields.



3. DAX Measures Configured

To support the report view visualizations, the following DAX measures were constructed:

- **Average Math Score:**

```
Avg Math = CALCULATE(AVERAGE('StudentsPerformance'[Score]), 'StudentsPerformance'[Subject] = "math score")
```

- **Average Reading Score:**

```
Avg Reading = CALCULATE(AVERAGE('StudentsPerformance'[Score]), 'StudentsPerformance'[Subject] = "reading score")
```

- **Average Writing Score:**

```
Avg Writing = CALCULATE(AVERAGE('StudentsPerformance'[Score]), 'StudentsPerformance'[Subject] = "writing score")
```

- **Overall Pass Percentage:**

```
Pass % = DIVIDE(CALCULATE(COUNTROWS('StudentsPerformance'), 'StudentsPerformance'[Pass/Fail] = "Pass"),
COUNTROWS('StudentsPerformance'), 0) * 100
```

4. Conclusion

In conclusion, the Week 7 data cleaning and transformation phase using Power BI Power Query was successfully completed. Raw data inconsistencies were thoroughly resolved through text trimming, capitalization, and index generation. Furthermore, essential performance metrics (Total, Average, Grade, and Pass/Fail) were engineered, and the subject scores were effectively unpivoted. This delivers a structured, error-free, and analysis-ready dataset that forms a solid foundation for building interactive dashboards and advanced DAX measures in Week 8.